

广义相对论（天文方向） 作业十一

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1. 将度规分解为 Minkowski 度规加上一个对称的微小扰动：

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

其中 $|h_{\mu\nu}| \ll 1$ 。证明在这种线性近似下：

$$\begin{aligned} R_{\mu\nu\rho}^{\sigma} &= \frac{1}{2}(\partial_{\nu}\partial_{\mu}h_{\rho}^{\sigma} + \partial_{\rho}\partial^{\sigma}h_{\mu\nu} - \partial_{\nu}\partial^{\sigma}h_{\mu\rho} - \partial_{\rho}\partial_{\mu}h_{\nu}^{\sigma}) \\ R_{\mu\nu} &= \frac{1}{2}(\partial_{\nu}\partial_{\rho}h_{\mu}^{\rho} + \partial_{\rho}\partial_{\mu}h_{\nu}^{\rho} - \partial_{\nu}\partial_{\mu}h - \square^2 h_{\mu\nu}) \\ R &= \partial_{\rho}\partial_{\mu}h^{\rho\mu} - \square^2 h \end{aligned}$$

从而进一步证明线性 Einstein 场方程为：

$$\partial_{\nu}\partial_{\mu}h + \square^2 h_{\mu\nu} - \partial_{\nu}\partial_{\rho}h_{\mu}^{\rho} - \partial_{\rho}\partial_{\mu}h_{\nu}^{\rho} - \eta_{\mu\nu}(\square^2 h - \partial_{\rho}\partial_{\sigma}h^{\sigma\rho}) = 2\kappa T_{\mu\nu}$$

在线性近似下：

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad g^{\mu\nu} \approx \eta^{\mu\nu} - h^{\mu\nu}$$

且 $\eta_{\mu\nu}$ 的导数为零：

$$\partial_{\rho}g_{\mu\nu} = \partial_{\rho}(\eta_{\mu\nu} + h_{\mu\nu}) = \partial_{\rho}h_{\mu\nu}$$

则首先计算微小扰动下的 Christoffel 符号：

$$\begin{aligned} \Gamma_{\mu\nu}^{\sigma} &= \frac{1}{2}g^{\sigma\rho}(\partial_{\mu}g_{\nu\rho} + \partial_{\nu}g_{\mu\rho} - \partial_{\rho}g_{\mu\nu}) \\ &\approx \frac{1}{2}(\eta^{\sigma\rho} - h^{\sigma\rho})[\partial_{\mu}(\eta_{\nu\rho} + h_{\nu\rho}) + \partial_{\nu}(\eta_{\mu\rho} + h_{\mu\rho}) - \partial_{\rho}(\eta_{\mu\nu} + h_{\mu\nu})] \\ &= \frac{1}{2}\eta^{\sigma\rho}(\partial_{\mu}h_{\nu\rho} + \partial_{\nu}h_{\mu\rho} - \partial_{\rho}h_{\mu\nu}) + \mathcal{O}(h^2) \\ &= \frac{1}{2}(\partial_{\mu}h_{\nu}^{\sigma} + \partial_{\nu}h_{\mu}^{\sigma} - \partial^{\sigma}h_{\mu\nu}) \end{aligned}$$

代入 Riemann 曲率张量的定义：

$$\begin{aligned} R_{\mu\nu\rho}^{\sigma} &= \partial_{\nu}\Gamma_{\mu\rho}^{\sigma} - \partial_{\rho}\Gamma_{\mu\nu}^{\sigma} \\ &= \partial_{\nu}\left[\frac{1}{2}(\partial_{\mu}h_{\rho}^{\sigma} + \partial_{\rho}h_{\mu}^{\sigma} - \partial^{\sigma}h_{\mu\rho})\right] - \partial_{\rho}\left[\frac{1}{2}(\partial_{\mu}h_{\nu}^{\sigma} + \partial_{\nu}h_{\mu}^{\sigma} - \partial^{\sigma}h_{\mu\nu})\right] \\ &= \frac{1}{2}[(\partial_{\nu}\partial_{\mu}h_{\rho}^{\sigma} + \partial_{\nu}\partial_{\rho}h_{\mu}^{\sigma} - \partial_{\nu}\partial^{\sigma}h_{\mu\rho}) - (\partial_{\rho}\partial_{\mu}h_{\nu}^{\sigma} + \partial_{\rho}\partial_{\nu}h_{\mu}^{\sigma} - \partial_{\rho}\partial^{\sigma}h_{\mu\nu})] \\ &= \frac{1}{2}(\partial_{\nu}\partial_{\mu}h_{\rho}^{\sigma} + \partial_{\rho}\partial^{\sigma}h_{\mu\nu} - \partial_{\nu}\partial^{\sigma}h_{\mu\rho} - \partial_{\rho}\partial_{\mu}h_{\nu}^{\sigma}) \end{aligned}$$

则：

$$R_{\mu\rho\nu}^{\sigma} = -R_{\mu\nu\rho}^{\sigma} = \frac{1}{2}(\partial_{\rho}\partial_{\mu}h_{\nu}^{\sigma} + \partial_{\nu}\partial^{\sigma}h_{\mu\rho} - \partial_{\rho}\partial^{\sigma}h_{\mu\nu} - \partial_{\nu}\partial_{\mu}h_{\rho}^{\sigma})$$

接下来，计算 Ricci 张量：

$$\begin{aligned} R_{\mu\nu} &= R_{\mu\sigma\nu}^{\sigma} \\ &= \frac{1}{2}(\partial_{\sigma}\partial_{\mu}h_{\nu}^{\sigma} + \partial_{\nu}\partial^{\sigma}h_{\mu\sigma} - \partial_{\sigma}\partial^{\sigma}h_{\mu\nu} - \partial_{\nu}\partial_{\mu}h_{\sigma}^{\sigma}) \\ &= \frac{1}{2}(\partial_{\rho}\partial_{\mu}h_{\nu}^{\rho} + \partial_{\rho}\partial_{\nu}h_{\mu}^{\rho} - \square^2 h_{\mu\nu} - \partial_{\nu}\partial_{\mu}h) \end{aligned}$$

此处的 $\square^2 = \partial_{\sigma}\partial^{\sigma}$ 为 d'Alembert 算符， $h = h_{\sigma}^{\sigma}$ 为 $h_{\mu\nu}$ 的迹。

最后，计算标量曲率：

$$\begin{aligned} R &= g^{\mu\nu}R_{\mu\nu} \\ &\approx \eta^{\mu\nu}R_{\mu\nu} \\ &= \frac{1}{2}\eta^{\mu\nu}(\partial_{\rho}\partial_{\mu}h_{\nu}^{\rho} + \partial_{\rho}\partial_{\nu}h_{\mu}^{\rho} - \square^2 h_{\mu\nu} - \partial_{\nu}\partial_{\mu}h) \\ &= \frac{1}{2}(\partial_{\rho}\partial_{\mu}h^{\rho\mu} + \partial_{\rho}\partial_{\nu}h^{\rho\nu} - \square^2 h - \partial^{\mu}\partial_{\mu}h) \\ &= \partial_{\rho}\partial_{\mu}h^{\rho\mu} - \square^2 h \end{aligned}$$

代入 Einstein 场方程：

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = -\kappa T_{\mu\nu}$$

得到线性 Einstein 场方程：

$$\frac{1}{2}(\partial_{\rho}\partial_{\mu}h_{\nu}^{\rho} + \partial_{\rho}\partial_{\nu}h_{\mu}^{\rho} - \square^2 h_{\mu\nu} - \partial_{\nu}\partial_{\mu}h) - \frac{1}{2}(\eta_{\mu\nu} + h_{\mu\nu})(\partial_{\rho}\partial_{\sigma}h^{\sigma\rho} - \square^2 h) = -\kappa T_{\mu\nu}$$

忽略高阶小量，整理得：

$$\partial_{\nu}\partial_{\mu}h + \square^2 h_{\mu\nu} - \partial_{\nu}\partial_{\rho}h_{\mu}^{\rho} - \partial_{\rho}\partial_{\mu}h_{\nu}^{\rho} - \eta_{\mu\nu}(\square^2 h - \partial_{\rho}\partial_{\sigma}h^{\sigma\rho}) = 2\kappa T_{\mu\nu}$$

证毕。